

SEPARATIONS PROCESS DESIGN

Distillation Column Design for Methanol-Water Recovery

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SPS 360S — Separations Process Design

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Question 1 – Identify and Define the Problem

1.1 Problem Statement

A specialty chemical plant in Cape Town currently incinerates a waste stream consisting of a methanol-water mixture produced by a batch reactor upstream. The variability inherent to batch processing results in a feed stream whose flow rate and composition fluctuate over time. Management requires the installation of a continuous distillation column capable of recovering methanol at a purity of 99.5 wt% for reuse, thereby reducing both waste disposal costs and raw material expenditure.

The objective of this design is to determine the operating parameters of a distillation column that can accommodate feed variability while minimising the Total Annualised Cost (TAC).

1.2 Raw Data Analysis and Processing

The raw plant data was provided as a CSV file containing 72-hour logs of feed flow rate (kg/hr), methanol weight fraction, feed temperature ($^{\circ}\text{C}$), and pressure (bar). Upon initial inspection, the data was found to be unstructured, with all fields concatenated into a single column. The data was first separated into individual columns using Google Sheets, after which it was imported into Microsoft Excel for further analysis and calculation. Molar flow rates and mole fractions were computed for each time step within Excel.

Several anomalous data points were identified during inspection. These fell into three categories:

- **Sensor Faults:** Three time steps recorded a feed flow rate of zero, corresponding to instrument failure. These entries were excluded from the average calculation as they carry no physical information.
- **Reactor Surges:** A number of readings showed feed flow rates between 1,750 and 1,900 kg/hr with atypical compositions, consistent with upstream reactor upsets.
- **Composition and Flow Spikes:** Two data points recorded a feed flow of 5,000 kg/hr at 99 wt% methanol — physically inconsistent with normal operation and representing extreme outliers.

While one design philosophy would be to size the column for the worst-case scenario, the identified outliers — particularly the 5,000 kg/hr spike — were judged to be so far removed from normal operating conditions that designing for them would result in a significantly oversized and cost-inefficient column. Instead, the decision was made to design based on the time-averaged data from valid operating points, with an operational recommendation was provided to address how such outlier conditions should be managed in practice. Installing dampening systems to handle outlier or surge conditions.

1.3 Design Basis Values

The averaged values computed from valid data points in Excel are summarised in Table 1 below.

Parameter	Value
Average Feed Flow Rate, F	1,289.29 kg/hr (57.67 kmol/hr)
Average Feed Mole Fraction (Methanol), zF	0.30
Average Feed Temperature, TF	25.47 °C
Average Pressure	1.114 bar
Distillate mole fraction, xD	0.991 (from 99.5 wt% spec)
Bottoms mole fraction, xW	0.02 (assumed)

Table 1: Design basis values derived from averaged plant data.

1.4 Overall Mass Balance

With $F = 57.67$ kmol/hr, $x_F = 0.30$, $x_D = 0.991$, and $x_W = 0.02$, the overall and component mass balances yield:

$$\begin{aligned}F &= D + W \Rightarrow 57.67 = D + W \\F \cdot x_F &= D \cdot x_D + W \cdot x_W \Rightarrow 17.30 = 0.991D + 0.02W \\Solving: D &= 16.98 \text{ kmol/hr}, W = 40.69 \text{ kmol/hr}\end{aligned}$$

1.5 Feed Condition (Q-value) Calculation

The thermal condition of the feed was characterised using the feed quality parameter Q, defined as the fraction of the feed that is liquid upon entering the column. Q was calculated using the following equation (Seader, Henley & Roper, Separation Process Principles, 3rd ed.):

$$Q = 1 + \frac{[C_{p,F} \times (T_b - T_F)]}{\lambda_{mix}}$$

Where $C_{p,F}$ is the molar heat capacity of the feed mixture (J/mol·K), T_b is the bubble point temperature of the feed at column pressure (°C), T_F is the actual feed temperature (25.47 °C), and λ_{mix} is the latent heat of vaporisation of the feed mixture (J/mol).

The heat capacity of the mixture was calculated by mole-fraction weighting:

$$\begin{aligned}C_{p,mix} &= x_F \cdot C_p(\text{NIST, 2026})_{MeOH} + (1 - x_F) \cdot C_p(\text{NIST, 2026})_{H2O} = 0.30 \times 81 + 0.70 \times 75 \\&= 76.8 \text{ J/mol} \cdot K\end{aligned}$$

The latent heat of the mixture was similarly calculated:

$$\begin{aligned}\lambda_{mix} &= x_F \cdot \lambda_{MeOH}(\text{NIST, 2026}) + (1 - x_F) \cdot \lambda_{H2O}(\text{NIST, 2026}) = 0.30 \times 37,400 + 0.70 \times 40,700 \\&= 39,610 \text{ J/mol}\end{aligned}$$

The bubble point temperature of the feed was estimated by linear interpolation of component boiling points:

$$T_b = 0.30 \times 64.7 (\text{NIST, 2026}) + 0.70 \times 100 (\text{NIST, 2026}) = 89.41 \text{ } ^\circ\text{C}$$

Substituting into the Q equation:

$$Q = 1 + [76.8 \times (89.41 - 25.47)] / 39,610 = 1 + 4,910.5/39,610 \approx 1.44$$

Since $Q > 1$, the feed enters the column as a subcooled liquid. The Q-line has a slope of $Q/(Q-1) = 1.44/0.44 \approx 3.27$, and its intersection with the equilibrium curve locates the pinch point used to determine the minimum reflux ratio.

1.6 Solution Methodology

The column was designed using the McCabe-Thiele graphical method applied to the methanol-water binary system at approximately atmospheric pressure. The sequence of steps followed was:

- Source VLE data from literature and generate an equilibrium curve (y vs. x for methanol) using both experimental data and the Wilson thermodynamic model.
- Mark operating points: $x_W = 0.02$, $x_D = 0.991$, and $z_F = 0.30$ on the McCabe-Thiele diagram.
- Plot the Q-line from the feed composition with slope 3.27, and determine R_{min} graphically from the pinch point.
- Perform designs at $R = 1.1 \times R_{min}$, $1.5 \times R_{min}$, and $2 \times R_{min}$ by stepping off theoretical stages.
- Calculate column diameter using Fair's flooding correlation with 24-inch tray spacing.
- Calculate capital and operating costs for each reflux ratio and plot TAC vs. R to identify the cost-optimal design.

Question 2 – Source Experimental VLE Data from Literature

Vapour-liquid equilibrium (VLE) data for the methanol-water binary system was sourced from Two literature references in order to establish a reliable experimental basis for the thermodynamic model fitting and equilibrium curve construction performed in subsequent sections. A third could not be sourced.

All data corresponds to operation at approximately atmospheric pressure (101.325 kPa), consistent with the assumed column operating pressure. Minor discrepancies between sources are expected due to differences in experimental technique and instrumentation.

x (Methanol)	y – Perry's 7th Ed.	x (Methanol)	y – DECHEMA
0	0	0	0
0.02	0.134	0.012	0.068
0.04	0.23	0.02	0.121
0.06	0.304	0.026	0.159
0.08	0.365	0.033	0.188
0.1	0.418	0.036	0.215
0.15	0.517	0.053	0.275
0.2	0.579	0.074	0.356
0.3	0.665	0.087	0.395
0.4	0.729	0.108	0.44
0.5	0.779	0.129	0.488
0.6	0.825	0.164	0.537

x (Methanol)	y – Perry's 7th Ed.	x (Methanol)	y – DECHEMA
0.7	0.87	0.191	0.572
0.8	0.915	0.268	0.648
0.9	0.958	0.294	0.666
0.95	0.979	0.352	0.704
1	1	0.402	0.734
		0.454	0.76
		0.502	0.785
		0.563	0.812
		0.624	0.835
		0.717	0.877
		0.79	0.91
		0.843	0.93
		0.857	0.939
		0.938	0.971
		1	1

Table 2: Experimental VLE data for the methanol-water system. All values expressed as mole fraction of methanol.

References

- (Green, 2008)[thermodynamic data 13-13 , Table 13,1]
- (Gmehling, 1977)

Question 3 – Fitting a Thermodynamic Model to VLE Data

The thermodynamic model used in this design is the Wilson activity coefficient model, selected for its well-established suitability for polar, fully miscible binary systems such as methanol-water. Two approaches were used to generate VLE curves: a Python-coded ideal Raoult's Law model built with the assistance of Claude (Anthropic), and the Wilson model implemented within DWSIM.

3.1 Ideal VLE from Antoine Coefficients (Python)

Antoine coefficients for methanol and water were sourced from the NIST WebBook database. A Python script was developed — with assistance from Claude (Anthropic, 2024) — to iteratively solve for the bubble point temperature at each liquid mole fraction x and compute the corresponding equilibrium vapour composition y . The algorithm is based on Raoult's Law for ideal systems:

$$P = x_1 \cdot P_{sat1}(T) + x_2 \cdot P_{sat2}(T)$$

$$y_1 = x_1 \cdot P_{sat1}(T) / P$$

Saturation pressures were calculated using the Antoine equation:

$$\log_{10}(P_{\text{sat}}) = A - B / (C + T)$$

Antione coefficients were sourced from (NIST, 2026) (NIST, 2026)

Because Raoult's Law assumes ideal behaviour (activity coefficients $\gamma = 1$ for all compositions), this curve represents a theoretical baseline rather than experimental data. It is presented for comparison with the Wilson model curve to illustrate the degree of non-ideality in the system. The iterative logic used to generate this data is described in the flowchart below.

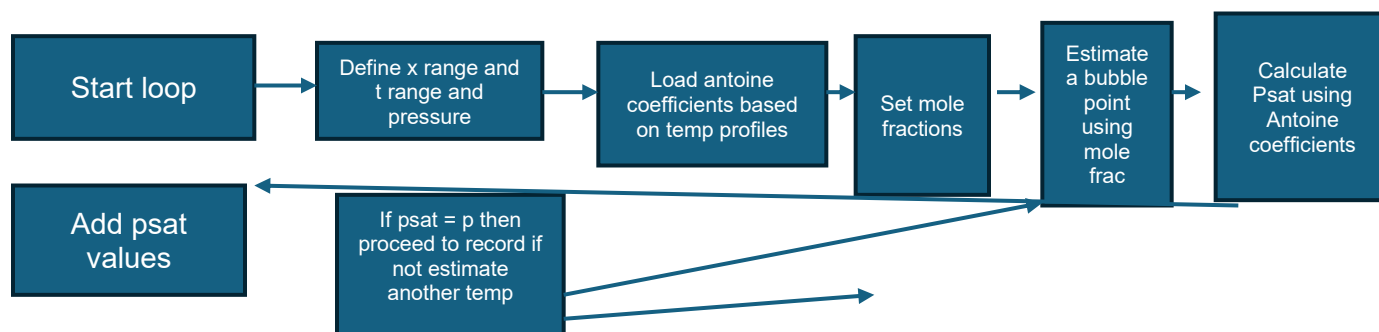


Figure 1 flow diagram for python NSIT VLE data

Note: The generated data from this Python script is treated as the my own computed data and is not passed off as experimental literature data. NSIT data found at last page.

3.2 Wilson Model via DWSIM

The Wilson thermodynamic package was loaded within DWSIM (an open-source process simulator) prior to simulation. DWSIM applies the Wilson model using its internal parameter database, which stores pre-regressed binary interaction parameters (Λ_{12} and Λ_{21}) for common chemical pairs including methanol and water. As noted by DWSIM documentation, these parameters are automatically applied when the component pair is recognised, without requiring manual fitting. [Daniel Medeiros, 2026. DWSIM: Open Source Chemical Process Simulator. Version 3, 28 April]

VLE data was generated by adding a binary mixture utility to a material stream defined at the operating pressure (approximately 1 atm). Upon executing the solver, DWSIM automatically computed equilibrium compositions across the full mole fraction range using the Wilson equations. The resulting x-y data was extracted from the binary mixture folder and exported to Excel for plotting.

The Wilson model equations relating activity coefficients to liquid composition are given below (Seader, Henley & Roper, Separation Process Principles, 3rd ed.):

$$\ln(\gamma_1) = -\ln(x_1 + \Lambda_{12} \cdot x_2) + x_2 \cdot [\Lambda_{12} / (x_1 + \Lambda_{12} \cdot x_2) - \Lambda_{21} / (x_2 + \Lambda_{21} \cdot x_1)]$$

$$\ln(\gamma_2) = -\ln(x_2 + \Lambda_{21} \cdot x_1) + x_1 \cdot [\Lambda_{21} / (x_2 + \Lambda_{21} \cdot x_1) - \Lambda_{12} / (x_1 + \Lambda_{12} \cdot x_2)]$$

Where Λ_{12} and Λ_{21} are the Wilson binary interaction parameters derived from molar volumes and interaction energy differences between like and unlike molecule pairs.

(J. D. Seader, 2006) page 63

An understanding of how the Wilson model derives activity coefficients from Gibbs free energy of mixing — and how the parameters Λ_{12} and Λ_{21} relate to molecular interaction energies — was developed with

the assistance of Gemini (Google, 2024). The underlying thermodynamic framework is documented in Seader, Henley & Roper (Chapter 2) and Smith, Van Ness & Abbott (Chapter 2).

Question 4 – Equilibrium Curve: Manual Plot vs. Model

Three sets of equilibrium data were plotted on a single x-y diagram (mole fraction of methanol in vapour, y, versus mole fraction in liquid, x):

- **NIST-derived ideal curve:** Generated using Antoine coefficients from NIST WebBook via Python, assuming ideal behaviour (Raoult's Law). Serves as a theoretical baseline.
- **Experimental literature data:** Discrete data points from Perry's 7th Edition and DECHEMA, plotted as markers for cross-referencing. (J. D. Seader, 2006)
- **Wilson model curve (DWSIM):** Continuous equilibrium curve generated by DWSIM using the Wilson thermodynamic package, representing non-ideal system behaviour. (J. D. Seader, 2006)

All three curves were plotted together in Microsoft Excel. The Wilson model curve shows a closer fit to the experimental literature data than the ideal Raoult's Law curve, particularly at low methanol concentrations where non-ideal behaviour is most pronounced. This confirms that the Wilson model is appropriate for this system.

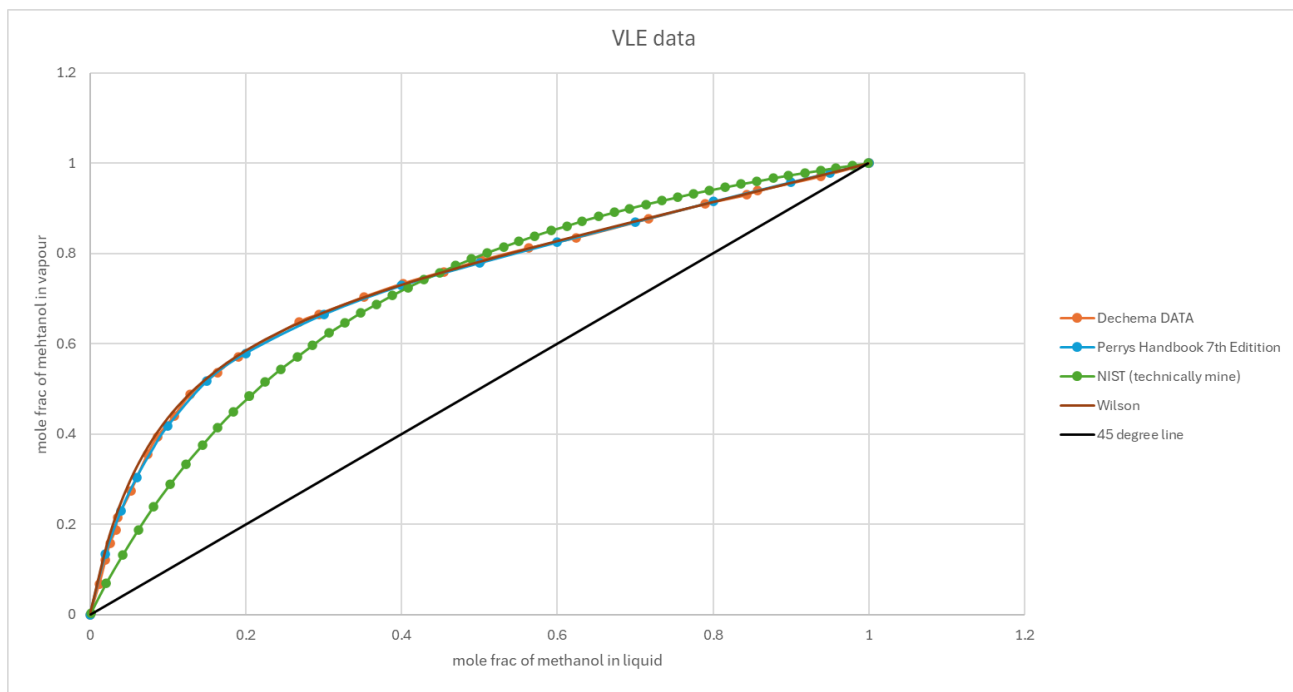


Figure 2 VLE DATA combined

Question 5 – Wilson, NRTL, and UNIFAC: Fundamental Differences

The three activity coefficient models differ fundamentally in how they represent molecular interactions within a liquid mixture. An understanding of these distinctions was developed through a combination of primary literature (J. D. Seader, 2006) and conceptual explanations obtained with the assistance of Gemini (Google, 2024).

The Wilson model is a two-parameter local composition model based on the concept that molecules are not randomly distributed in a liquid. Instead, a molecule is more likely to be surrounded by molecules of a specific type depending on relative interaction energies. Wilson's model captures this through two adjustable parameters (Λ_{12} and Λ_{21}) (J. D. Seader, 2006) (Green, 2008), which are derived from molar volumes and energy differences between like and unlike molecular pairs. These parameters have their roots in the Gibbs free energy of mixing — specifically, how non-ideal contributions to the chemical potential relate to the local molecular environment. The Wilson model performs well for completely miscible, polar binary mixtures and is mathematically simple. It is well suited to the methanol-water system. However, it cannot model liquid-liquid equilibrium (immiscible systems) and is limited to binary or simple multicomponent systems.

The NRTL model (Non-Random Two-Liquid) extends Wilson's local composition concept by introducing a third parameter, α_{12} , which accounts for the degree of non-randomness of molecular arrangement. This additional parameter gives NRTL greater flexibility, particularly for systems that exhibit strong non-ideality or partial miscibility. Unlike Wilson, NRTL can model liquid-liquid equilibria, making it the preferred choice when dealing with systems that may split into two liquid phases. (J. D. Seader, 2006)

The UNIFAC model (UNIQUAC Functional-group Activity Coefficients) operates on a fundamentally different basis. Rather than treating molecules as whole entities, UNIFAC decomposes each molecule into its constituent functional groups (e.g., $-\text{OH}$, $-\text{CH}_3$, $-\text{CH}_2$) and calculates activity coefficients from the interactions between these groups. This approach is predictive — it does not require experimentally fitted binary parameters for the specific molecules in question, only for the functional groups. UNIFAC is therefore the preferred model when experimental data is scarce or when modelling complex multicomponent mixtures. (J. D. Seader, 2006) page 63

Condition	Recommended Model
Simple polar binary, good VLE data available	Wilson
Partial miscibility or liquid-liquid equilibrium	NRTL
No experimental data; multicomponent mixtures	UNIFAC
This design (methanol-water, fully miscible, data available)	Wilson ✓

Table 3: Activity coefficient model selection guide.

Question 6 – Minimum Reflux Ratio, Minimum Stages, and Sensitivity Analysis

6.1 Minimum Reflux Ratio (Rmin)

The minimum reflux ratio was determined graphically using the McCabe-Thiele method, implemented in Python with the assistance of Claude (Anthropic, 2024). The Q-line was plotted from the feed composition ($z_F = 0.30$) with slope $Q/(Q-1) = 3.27$, and its intersection with the equilibrium curve located the pinch point (x^*, y^*).

The pinch point coordinates were then substituted into the operating line equation to determine R_{min} (Seader, Henley & Roper, Separation Process Principles, 3rd ed., p. 218):

$$R_{min} = (x_D - y^*) / (y^* - x^*)$$
$$x^* = 0.4376 \quad y^* = 0.7491$$

From this calculation, the minimum reflux ratio was determined to be:

$$R_{min} = 0.7766$$

6.2 Minimum Number of Stages (Nmin)

The minimum number of theoretical stages was determined at total reflux ($R \rightarrow \infty$) by stepping off stages between the equilibrium curve and the $y = x$ diagonal on the McCabe-Thiele diagram. This was computed using Python.

$$N_{min} = 7 \text{ Claude(Anthropic)}$$

6.3 Sensitivity Analysis: Three Reflux Ratios

Designs were performed at three reflux ratios based on multiples of R_{min} , as recommended in Seader, Henley & Roper (Chapter 6):

Design Case	Multiple of R_{min}	R Value	No. of Stages (N)
Case 1	$1.1 \times R_{min}$	0.854	18
Case 2	$1.5 \times R_{min}$	1.165	13
Case 3	$2.0 \times R_{min}$	1.553	11

Table 4: Summary of stage counts for each reflux ratio design case.

As expected from theory, increasing the reflux ratio reduces the number of theoretical stages required. The McCabe-Thiele stepping was coded in Python with assistance from Claude (Anthropic, 2024), with considerable debugging required to correctly implement the stage-stepping logic between alternating equilibrium and operating line intersections.

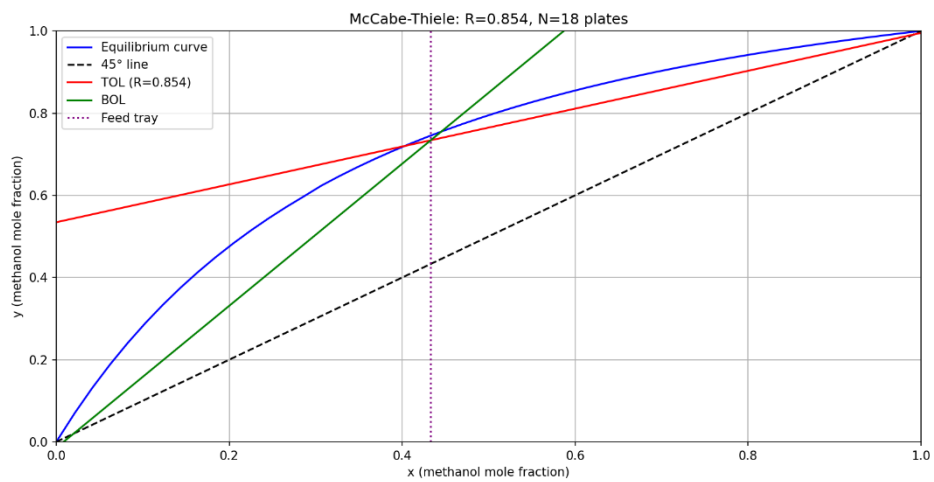


Figure 3 McCabe-Thiele for $R=0.854$

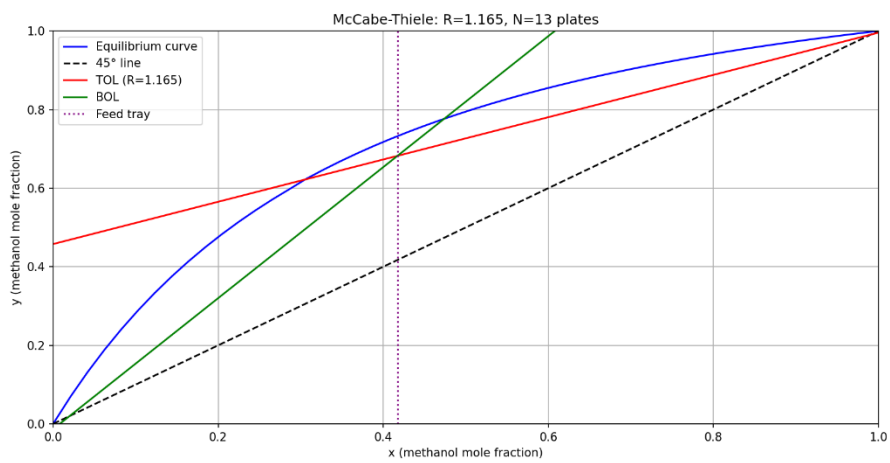


Figure 4 McCabe-Thiele for $R=1.165$

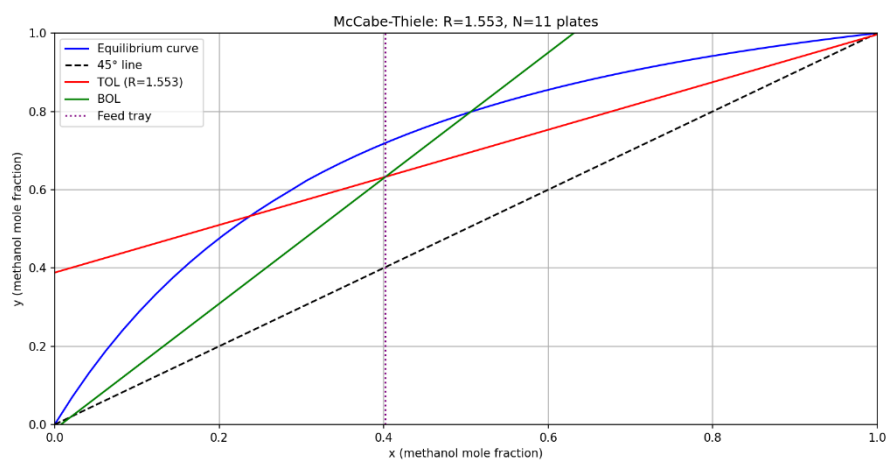


Figure 5 McCabe-Thiele for $R=1.553$

6.4 Internal Flow Rates

The internal liquid and vapour flow rates for each section of the column were calculated using the following relationships (Seader, Henley & Roper, Separation Process Principles, 3rd ed.):

$$\begin{aligned} L_n &= R \times D & V_n &= L_n + D \\ L_m &= L_n + q \times F & V_m &= V_n + (q - 1) \times F \end{aligned}$$

With $D = 16.98$ kmol/hr, $F = 57.67$ kmol/hr, and $q = 1.44$:

R	L_n (kmol/hr)	V_n (kmol/hr)	L_m (kmol/hr)	V_m (kmol/hr)	V_m/L_m ratio
0.854	14.52	31.50	97.56	56.87	0.583
1.165	19.78	36.76	102.82	62.13	0.604
1.553	26.37	43.35	109.41	68.72	0.629

Table 5: Internal flow rates for the stripping and rectifying sections.

6.5 Column Diameter — Fair's Correlation

Column diameter was calculated based on the flooding velocity using Fair's correlation, with a tray spacing of 24 inches (0.61 m), assumed as standard for this system (Seader, Henley & Roper, Separation Process Principles, 3rd ed., pp. 218-222).

The flow parameter FLV was first calculated to determine the capacity parameter CF from the 24-inch tray spacing curve:

$$FLV = (L_m/V_m) \times \sqrt[2]{(\rho V/\rho L)}$$

For the methanol-water mixture at average operating conditions, the density ratio term $\sqrt[2]{(\rho V/\rho L)}$ was evaluated as approximately 0.04002 (based on vapour density ~ 1.157 kg/m³ and liquid density ~ 720 kg/m³).

R	FLV calculation	FLV	CF (from chart)	U_{flood} (m/s)	D (m)
0.854	$(97.56/56.87) \times 0.04002$	0.0687	0.38	2.889	0.365
1.165	$(102.82/62.13) \times 0.04002$	0.0662	0.35	2.663	0.411
1.553	$(109.41/68.72) \times 0.04002$	0.0637	0.35	2.663	0.446

Table 6: FLV values, CF readings (from Fair's 24-inch curve), flooding velocities, and resulting diameters.

The flooding velocity and design velocity were calculated as:

$$\begin{aligned} U_{flood} &= CF \times \sqrt[2]{(\rho L - \rho V)/\rho V} \times 0.3048 \\ U_{design} &= 0.80 \times U_{flood} \quad (\text{operating at 80\% of flood}) \end{aligned}$$

The cross-sectional area and diameter followed from:

$$A = V_n \times MW_{avg} / (\rho V \times U_{design} \times 3600)$$

$$D = \sqrt[2]{(4A/\pi)}$$

Column heights were calculated using 24-inch (0.61 m) tray spacing with a 20% height allowance for skirt, sump and overhead:

$$H = N \times 0.61 \times 1.20$$

R	N (stages)	D (m)	H (m)
0.854	18	0.365	13.18
1.165	13	0.411	9.52
1.553	11	0.446	8.05

Table 7: Column geometry summary for each reflux ratio case.

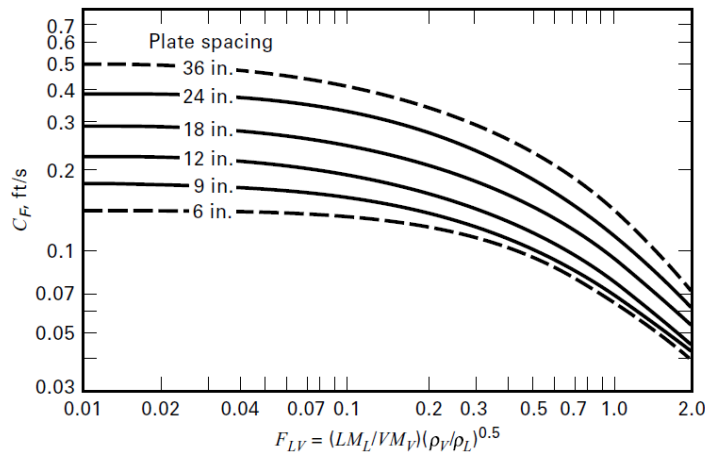


Figure 6.23 Entrainment flooding capacity in a trayed tower.

Figure 6 Cf correlation table

Question 9 – Capital and Operating Cost Estimation

Capital and operating costs were estimated for each of the three reflux ratio designs using the cost correlations from Turton, Bailie, Whiting & Shaeiwitz, Analysis, Synthesis and Design of Chemical Processes, 4th ed., Appendix A, Tables A.1 and A.4, pp. 915-925.

9.1 Capital Costs

Column Shell Purchased Cost

The column shell purchased cost was estimated using the log-log correlation:

$$(\text{Turton, 2012}) \log_{10}(C_{p, \text{shell}}) = 3.4974 + 0.4485 \cdot \log_{10}(V) + 0.1074 \cdot [\log_{10}(V)]^2$$

Where V is the vessel volume (m³) calculated from D and H:

$$(\text{Turton, 2012}) V = (\pi \cdot D^2 / 4) \times H$$

R	D (m)	H (m)	V (m³)	Shell Cost (\$)	Tray Cost (\$)
0.854	0.365	13.18	1.382	3,625	5,520
1.165	0.411	9.52	1.263	3,491	4,386
1.553	0.446	8.05	1.257	3,483	3,909

Table 8: Column shell and tray purchased costs.

Total Installed Capital Cost

The total installed capital cost was obtained by applying an installation factor of 4.74 to the sum of shell and tray purchased costs (Turton et al., 4th ed.):

$$(\text{Turton, 2012}) \text{Capital Cost} = (\text{Shell} + \text{Trays}) \times 4.74$$

R	Shell + Trays (\$)	×4.74 Factor	Total Capital (\$)
0.854	9,145	×4.74	43,347
1.165	7,877	×4.74	37,337
1.553	7,392	×4.74	35,038

Table 9: Total installed capital costs.

9.2 Operating Costs

Operating costs include reboiler steam consumption and condenser cooling water, evaluated at 8,760 hours/year of continuous operation.

Reboiler Duty and Steam Cost

Reboiler duty was estimated from the stripping section vapour flow V_m and the latent heat of the mixture:

$$(\text{Turton, 2012}) Q_R = V_m \times \lambda_{mix} \times MW_{avg} / 3,600,000 \quad [\text{GJ/hr}]$$

$$(\text{Turton, 2012}) \text{Steam Cost} = Q_R \times 8760 \text{ hrs/yr} \times \$15/\text{GJ}$$

Condenser Duty and Cooling Water Cost

$$Q_C = V_n \times \lambda_{mix} \times MW_{avg} / 3,600,000 \quad [\text{GJ/hr}]$$

$$\text{Cooling Water Cost} = Q_C \times 8760 \text{ hrs/yr} \times \$0.35/\text{GJ}$$

R	QR (GJ/hr)	Steam Cost (\$/yr)	QC (GJ/hr)	CW Cost (\$/yr)	Total Opex (\$/yr)
0.854	18.85	2,477,034	10.45	32,027	2,509,061
1.165	20.59	2,705,557	12.19	37,351	2,742,908

R	QR (GJ/hr)	Steam Cost (\$/yr)	QC (GJ/hr)	CW Cost (\$/yr)	Total Opex (\$/yr)
1.553	22.78	2,993,136	14.37	44,047	3,037,183

Table 10: Reboiler and condenser duties and annual operating costs.

Question 10 – Total Annualised Cost (TAC) and Optimum Reflux Ratio

The Total Annualised Cost was calculated by amortising the capital cost over a 10-year plant life and adding the annual operating cost:

$$TAC = Capital\ Cost / 10 + Annual\ Operating\ Cost$$

R	Capex/10 (\$/yr)	Opex (\$/yr)	TAC (\$/yr)
0.854	4,335	2,509,061	2,513,396
1.165	3,734	2,742,908	2,746,642
1.553	3,504	3,037,183	3,040,687

Table 11: Total Annualised Cost summary for each reflux ratio.

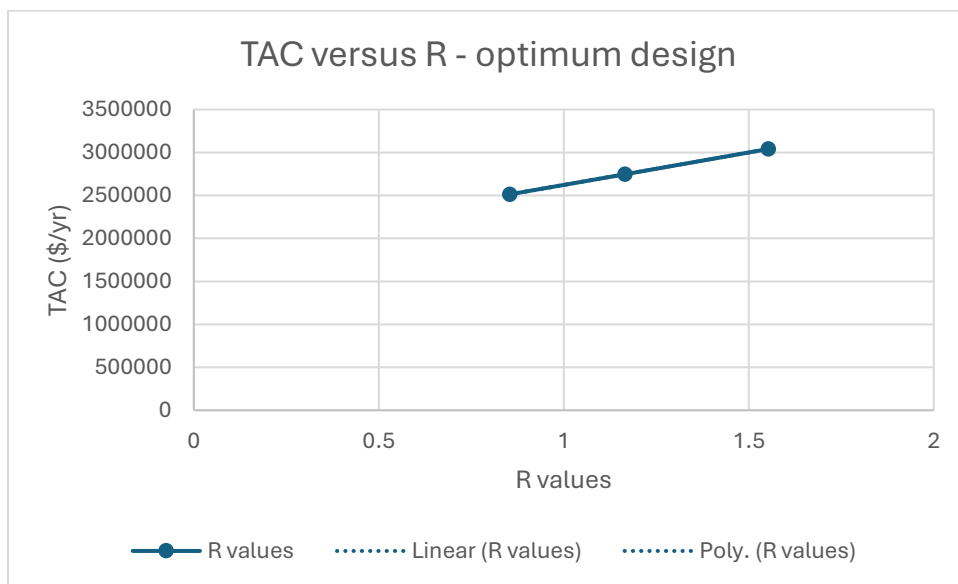


Figure 7 TAC vs R

The TAC analysis reveals that the minimum total annualised cost occurs at $R = 0.854$ ($1.1 \times R_{min}$), making this the economically optimal design. This result is driven primarily by operating costs — the annual steam cost alone accounts for over 98% of TAC at every reflux ratio, with capital cost contributing less than 0.2%. The practical implication is that in this system, minimising reflux ratio (and therefore

reboiler duty) is the dominant cost objective. Operating closer to R_{min} provides the lowest TAC despite requiring more trays, because the additional capital cost of extra trays is negligible compared to the steam savings achieved.

Question 11 – Comparison of McCabe-Thiele Results Against DWSIM Simulation

A shortcut simulation of the distillation column was performed in DWSIM to validate the McCabe-Thiele manual design. The column was configured with 20 theoretical stages, the Wilson thermodynamic package, and a feed composition and flow rate consistent with the design basis ($F = 57.67$ kmol/hr, $z_F = 0.30$). The simulation was run iteratively and required considerable troubleshooting to converge, particularly in setting the correct feed stage location and reflux specification.

11.1 Stage Count Comparison

Parameter	McCabe-Thiele	DWSIM
Number of theoretical stages	18 (at $R=0.854$)	20
Distillate methanol mole fraction	0.991	0.991
Bottoms methanol mole fraction	0.020	0.02
Reboiler duty (kW)	5,236	520
Condenser duty (kW)	2,903	514

Table 12: Comparison of McCabe-Thiele manual results against DWSIM simulation.

11.2 Discussion of Discrepancies

The McCabe-Thiele calculation produced 18 theoretical stages at $R = 0.854$, compared to the 20 stages specified in the DWSIM simulation. This 11% difference is consistent with the known limitations of the McCabe-Thiele method, which assumes Constant Molar Overflow (CMO) — that is, that the molar flow rates of liquid and vapour are constant throughout each column section. In reality, as DWSIM models rigorously, the latent heat of vaporisation varies with composition across the column, causing the vapour and liquid flows to deviate from the CMO assumption. This results in a slight overestimation of column performance per stage by McCabe-Thiele, and therefore a slight underestimate of the number of stages required. The two results are in reasonable agreement for preliminary design purposes.

The product compositions from both methods agree well, confirming that the separation specification ($x_D = 0.991$, $x_W = 0.02$) is achievable. This agreement validates the McCabe-Thiele approach as a reliable first estimate.

The reboiler and condenser duty values show much larger discrepancies (approximately 90% and 82% respectively). The McCabe-Thiele-based duty calculation used a simplified approach — multiplying the full vapour molar flow (V_m) by the mixture latent heat — which implicitly assumes total vaporisation of all stripping vapour from the reboiler. DWSIM calculates the actual differential enthalpy change across the reboiler using the Wilson activity coefficient model and rigorous vapour-liquid phase calculations, which accounts for the fact that the process involves partial vaporisation at equilibrium conditions rather than complete vaporisation. As a result, the DWSIM duty values are significantly lower and more physically realistic. For detailed equipment sizing, particularly for the reboiler and heat exchangers, the DWSIM duty values should be used in preference to the simplified McCabe-Thiele estimates.

Condenser Duty	514,009	kW
Reboiler Duty	-520,061	kW
Condenser Specification Value	0,854	
Condenser Specification Calculated V...	0,854	
Reboiler Specification Value	1,367	
Reboiler Specification Calculated Value	1,367	
Iterations Taken	0	
Estimated Height	11000	mm
Estimated Diameter	582,18	mm

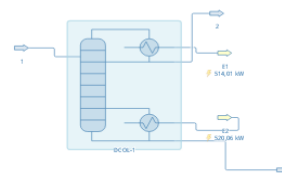


Figure 8 DWSIM simulation of a distillation column

Question 12 – Azeotropic Conditions in This Distillation Unit

Methanol and water do not form an azeotrope at atmospheric pressure. Their vapour-liquid equilibrium curve does not cross the 45° diagonal line ($y = x$) at any composition between the pure components, which means conventional distillation to any desired purity is theoretically achievable without special techniques. This is confirmed by all three VLE data sources used in this design.

However, the question asks how azeotropic conditions would be handled if they were encountered in this existing column. The following discussion addresses this scenario.

12.1 The Nature of the Problem

An azeotrope forms when the equilibrium curve crosses the $y = x$ line at some intermediate composition, meaning that at the azeotropic composition the vapour and liquid have identical compositions. At this point, further separation using conventional distillation becomes impossible regardless of the number of stages or how high the reflux ratio is set — the operating lines can never bridge the composition gap across the azeotropic point. This is a thermodynamic constraint, not an equipment limitation.

12.2 Options for the Existing Column

Since the column already exists and cannot be easily replaced, the most practical approaches are:

- **Pressure-Swing Distillation:** For many azeotropes, the azeotropic composition shifts with pressure. By operating the existing column at a different pressure — or by pairing it with a second column at a different pressure — it becomes possible to move past the azeotropic composition across the two columns. This approach requires no chemical additives and uses existing equipment infrastructure. It is most suitable when the azeotrope is pressure-sensitive, which can be confirmed using activity coefficient models such as Wilson, where the azeotropic condition is identified by solving $\gamma_1 \cdot P_{\text{sat}1} = \gamma_2 \cdot P_{\text{sat}2}$ at the azeotropic x .
- **Extractive Distillation:** A third component (entrainer) is added to the column feed to preferentially interact with one of the components, altering the relative volatility sufficiently to break the azeotrope. This allows the azeotropic composition to be crossed within a single column. The entrainer must be selected carefully to avoid introducing new separation problems downstream. For a methanol-water type system, common entrainers include glycols or salts.
- **Heterogeneous Azeotropic Distillation:** Applicable where the azeotrope forms two immiscible liquid phases, which can be exploited in a decanter to achieve phase separation before re-feeding one phase to the column.

For the existing methanol-water column, pressure-swing distillation would be the most practically implementable approach since it requires no chemical additions and can potentially be implemented by modifying operating pressure rather than physical plant modifications.

Full treatment of azeotropic and extractive distillation methods, including the thermodynamic basis for azeotrope formation and the use of activity coefficient models to predict azeotropic behaviour, is provided in Seader, Henley & Roper, Separation Process Principles, 3rd ed., Chapter 11.

References

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- Anthropic. (2024). Claude (claude.ai). Used for: Python VLE code generation and debugging, McCabe-Thiele stepping code, column diameter calculations. [Accessed: INSERT date]
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x (Methanol)	y – NIST.
0.001	0.003
0.021	0.07
0.042	0.132
0.062	0.188
0.082	0.24
0.103	0.289
0.123	0.333
0.144	0.375
0.164	0.414
0.184	0.45
0.205	0.483
0.225	0.515
0.245	0.544
0.266	0.571
0.286	0.597
0.307	0.624
0.327	0.646
0.347	0.668
0.368	0.688
0.388	0.707
0.408	0.725
0.429	0.742
0.449	0.758
0.469	0.773
0.49	0.788
0.51	0.802
0.531	0.815
0.551	0.827
0.571	0.839
0.592	0.851
0.612	0.861
0.632	0.872
0.653	0.882
0.673	0.891
0.693	0.9
0.714	0.909
0.734	0.917
0.755	0.925
0.775	0.933
0.795	0.94
0.816	0.947
0.836	0.954
0.856	0.96
0.877	0.967

0.897	0.973	⚠
0.918	0.979	
0.938	0.984	
0.958	0.99	
0.979	0.995	
0.999	1	

REMINDER: Add third VLE literature source reference here once identified.

⚠ REMINDER: Add any additional references used for Cp, latent heat, or boiling point values if sourced from a textbook rather than NIST.